

A STRICT CERTIFIED SEARCH EXTENSION FOR ERDŐS PROBLEM 647

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ABSTRACT. Let $\tau(m)$ denote the number of positive divisors of m . Erdős Problem 647 asks whether there is an $n > 24$ for which $\max_{m < n} (m + \tau(m)) \leq n + 2$. We give a reproducible finite exclusion for

$$9,174,471,185,880,000,000 < n \leq 9,180,628,549,092,000,000.$$

It reconstructs all 96 residue classes in the published theorem-grade modular reduction, expands them into 50,784 arithmetic progressions, and checks 5,078,450,784 progression cells. Exact small-prime divisor lower bounds reject all but 21 cells; those 21 are factored independently and each violates one of the shifts $1 \leq k \leq 6$. Combined with the reported previous computation through the left endpoint, this gives a strict new computational frontier at 9,180,628,549,092,000,000. The inherited prefix was not rerun here, and the global existence problem remains open.

1. INTRODUCTION

For a positive integer m , write $\tau(m)$ for its number of positive divisors. The problem, originating with Erdős and Selfridge, asks whether there is an integer $n > 24$ such that

$$(1) \quad \max_{1 \leq m < n} (m + \tau(m)) \leq n + 2.$$

The value 24 is the last known equality case [2, 1]. Direct and certificate-based finite computations have successively treated ranges through 10^{12} and beyond [5, 6, 7], but no finite computation alone settles the existential question. Erdős also noted that a related fixed-window statement follows from Schinzel’s Hypothesis H [3]; that conditional statement does not control the full prefix in (1).

The public program of Tordjman reports exclusion through

$$9,174,471,185,880,000,000$$

[7]. The contribution here is an independently replayed interval above that endpoint. The distinction is important: Theorem 3.1 is supported by the new local certificate, whereas Corollary 3.2 additionally inherits the earlier computation.

2. LOCAL AND MODULAR REDUCTIONS

Putting $k = n - m$ transforms (1), for a fixed n , into the equivalent family

$$(2) \quad \tau(n - k) \leq k + 2 \quad (1 \leq k < n).$$

Thus a single exact inequality $\tau(n - k) > k + 2$ rejects n .

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We use the following published prerequisite. Its proof path is formalized in Lean 4 and, at the stage used here, does not use the later open-residue axiom [4].

Proposition 2.1 (Hughes modular reduction). *If $n > 84$ satisfies (2), then $n = 2520N$ for an integer N , and $r = N \bmod M$ belongs to*

$$(3) \quad \mathcal{R} = \{0 \leq r < M : dr \not\equiv 1 \pmod{q} \text{ for every } d \in \mathcal{D}, q \in \mathcal{Q}\},$$

where

$$M = 46189 = 11 \cdot 13 \cdot 17 \cdot 19,$$

$$\mathcal{Q} = \{11, 13, 17, 19\},$$

$$\mathcal{D} = \{105, 140, 210, 252, 280, 315, 420, 504, 630, 840, 1260, 2520\}.$$

Moreover, $|\mathcal{R}| = 96$.

We deliberately retain all 96 classes. In particular, the present result does not use the subsequent split into 55 closed and 41 open classes, nor the problem-specific axiom associated with that split.

Set

$$A = 2520 \cdot 46189 \cdot 529 = 61,573,632,120,$$

$$u_0 = 149,000,000, \quad u_1 = 149,100,000.$$

For $r \in \mathcal{R}$ and $0 \leq s < 529$, define

$$(4) \quad B_{r,s} = 2520(46189s + r), \quad n = Au + B_{r,s}.$$

Every N with $N \bmod M \in \mathcal{R}$ has a unique representation of this form. Since $0 \leq B_{r,s} < A$, every candidate in $(Au_0, Au_1]$ occurs among the cells with $u_0 \leq u \leq u_1$. This inclusive choice of u_0 is essential for covering the partial layer immediately to the right of Au_0 ; cells with $u = u_1$ and $B_{r,s} > 0$ are harmless overcoverage.

3. THE CERTIFIED INCREMENT

Theorem 3.1. *No integer n with*

$$9,174,471,185,880,000,000 < n \leq 9,180,628,549,092,000,000$$

satisfies (1).

Proof. The endpoint identities are

$$Au_0 = 9,174,471,185,880,000,000, \quad Au_1 = 9,180,628,549,092,000,000.$$

Since the interval lies above 84, Proposition 2.1 and the parameterization (4) include every possible candidate among the

$$96 \cdot 529 \cdot (u_1 - u_0 + 1) = 5,078,450,784$$

enumerated cells.

For each cell the verifier examines $F_k = n - k$ for $1 \leq k \leq 16$. It removes the exact valuations of every prime $p \leq 100,000$. If $Q = \prod p^{v_p(F_k)}$ is the removed part and $C = F_k/Q$, then $\gcd(Q, C) = 1$ and hence

$$(5) \quad \tau(F_k) = \tau(Q)\tau(C) \geq \begin{cases} \tau(Q), & C = 1, \\ 2\tau(Q), & C > 1. \end{cases}$$

The implementation rejects a cell only when this integer lower bound exceeds $k + 2$. It rejects 5,078,450,763 cells in this way.

The remaining 21 cells are emitted rather than accepted. A separate Python verifier factors F_k from scratch using deterministic 64-bit primality testing and Pollard–rho factorization, checks the product and every prime factor, reconstructs $\tau(F_k)$, and finds $\tau(F_k) > k + 2$ in each case. The killing-shift distribution is

k	1	2	3	4	6
hard cells killed	14	3	2	1	1.

There are no unresolved cells. Thus every cell violates (2), and consequently no n in the stated interval satisfies (1). $\square$

Corollary 3.2 (reported combined frontier). *Combining Theorem 3.1 with the previously reported exclusion through 9,174,471,185,880,000,000 [7] gives a reported combined computational frontier through*

$$9,180,628,549,092,000,000.$$

Remark 3.3. The wording of Corollary 3.2 records a trust split. The current artifact independently replays the entire strict increment, including its left partial layer. It audits the source, logs, and exceptional factorizations of the inherited computation, but it does not rerun that full earlier scan.

4. CERTIFICATE AND INDEPENDENT REPLAY

The serialized input records the endpoints, the formula (4), the full Cartesian product $\mathcal{R} \times \{0, \dots, 528\}$, and cryptographic hashes. The verifier does not accept a copied residue list: it regenerates $\mathcal{R}$ from (3), proves that its size is 96, and checks that the 50,784 rows are exactly the required Cartesian product. It also checks the identities defining A and the two endpoints and refuses arithmetic outside the signed 64-bit domain.

The decisive output is

Quantity	Exact value
Residues r	96
Subprogressions (r, s)	50,784
Values of u per subprogression	100,001
Enumerated cells	5,078,450,784
Strict lower-bound kills	5,078,450,763
Independently factored hard cells	21
Unresolved cells	0

As one representative hard certificate, for

$$n = 9,175,022,326,300,168,800, \quad k = 6,$$

the verifier reconstructs

$$n - k = 2 \cdot 3 \cdot 176017 \cdot 8687628966047, \quad \tau(n - k) = 16 > 8 = k + 2.$$

All 21 exact records, including (r, s, u, n, k) , the prime factorization, and $\tau(n - k)$, are stored in `certificates/finite_extension_certificate.json`.

From the project root, the decisive reconstruction command is

```
python3 verification/verify_extension.py --threads 12
```

The certificate’s SHA-256 digest is

$$19f8f31a1d037ddae404448ba13863f10607a23f24414b57628bf13b6ecc83c3.$$

The verifier also binds the input, pair table, C source, Python source, and discovery log by SHA-256.

5. SCOPE AND DISCLOSURE

This result is a strict finite bound, not a proof that a second n exists or does not exist. Large unsuccessful searches cannot decide the remaining infinite tail. A second novelty search on August 29, 2026 found no publication or database entry reporting the exact endpoint above; this is a bounded search statement, not a claim about unpublished work.

The new replay used no proof assistant. Its imported modular prerequisite was formalized in Lean 4 by Hughes; the present project inspected that theorem boundary but did not invoke Lean locally. OpenAI Codex assisted with literature search, proof exploration, code, tests, and prose. The discovery scan used an existing C program [7]; the decisive replay was separately implemented and used exact integer arithmetic. Runs were deterministic and used no random seed.

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